

Homework 3

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① $y^{(5)} - y' = 0$

Ⓐ Characteristic Equation: $\lambda^5 - \lambda = 0$

$$\lambda(\lambda^4 - 1) = \lambda(\lambda^2 - 1)(\lambda^2 + 1) = \lambda(\lambda - 1)(\lambda + 1)(\lambda^2 + 1) = 0$$

$$\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = -1, \lambda_4 = -i, \lambda_5 = +i$$

$$y(x) = A + B \cdot e^x + C \cdot e^{-x} + D \cdot e^{-ix} + E \cdot e^{ix}$$

$$y(x) = A + B \cdot e^x + C \cdot e^{-x} + D \cdot \cos x + E \cdot \sin x //$$

Ⓑ $y(x) = A + B \cdot e^x + C \cdot e^{-x} + D \cdot \cos x + E \cdot \sin x \Rightarrow y(0) = A + B + C + D = 0$

$$y'(x) = B \cdot e^x - C \cdot e^{-x} - D \cdot \sin x + E \cdot \cos x \Rightarrow y'(0) = B - C + E = 1$$

$$y''(x) = B \cdot e^x + C \cdot e^{-x} - D \cdot \cos x - E \cdot \sin x \Rightarrow y''(0) = B + C - D = 0$$

$$y'''(x) = B \cdot e^x - C \cdot e^{-x} + D \cdot \sin x - E \cdot \cos x \Rightarrow y'''(0) = B - C - E = 0$$

$$y^{(4)}(x) = B \cdot e^x + C \cdot e^{-x} + D \cdot \cos x + E \cdot \sin x \Rightarrow y^{(4)}(0) = B + C + D = 0$$

Therefore; $A = 0, B = \frac{1}{4}, C = -\frac{1}{4}, D = 0, E = \frac{1}{2}$

So; $y(x) = \frac{1}{4} \cdot e^x - \frac{1}{4} e^{-x} + \frac{1}{2} \sin x //$

② $y''' - 3y'' + 3y' - y = 0$

Ⓐ Characteristic Equation: $\lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$

$$(\lambda - 1)^3 = 0$$

$\lambda = 1$ with multiplicity 3.

$$y(x) = (A + Bx + Cx^2)e^x //$$

Ⓑ $y(x) = (A + Bx + Cx^2)e^x \Rightarrow y(0) = A = 0$

$$y'(x) = (B + C \cdot 2x)e^x + (A + Bx + Cx^2)e^x \Rightarrow y'(0) = B + A = 1$$

$$y''(x) = 2C \cdot e^x + (B + C \cdot 2x)e^x + (B + C \cdot 2x)e^x + (A + Bx + Cx^2)e^x \Rightarrow y''(0) = 2C + B + B + A = 0$$

Therefore; $A = 0, B = 1, C = -1$

So; $y(x) = (x - x^2) \cdot e^x //$

③ $x^2 y'' - 3xy' + 4y = 0$

Ⓐ $x = e^t \Rightarrow t = \ln x$

$$Y(t) = y(e^t) = y(x)$$

$$\frac{dy}{dx} = \frac{dY}{dt} \cdot \frac{dt}{dx} = y' \cdot \frac{1}{x}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left[y' \cdot \frac{1}{x} \right] = \frac{dY}{dt} \cdot \frac{dt}{dx} \cdot \frac{1}{x} + y' \cdot \frac{d}{dx} \left[\frac{1}{x} \right] = y'' \cdot \frac{1}{x^2} - y' \cdot \frac{1}{x^2} = \frac{(y'' - y')}{x^2}$$

$$x^2 \frac{(y'' - y')}{x^2} - 3x \cdot \frac{y'}{x} + 4y = 0$$

$$y'' - y' - 3y' + 4y = 0$$

$$y'' - 4y' + 4y = 0$$

Characteristic Equation: $\lambda^2 - 4\lambda + 4 = 0$

$$(\lambda - 2)^2 = 0$$

$\lambda = 2$ with multiplicity 2.

$$Y(t) = (A + Bt)e^{2t}$$

$$y(x) = (A + B \cdot \ln x) \cdot e^{2 \ln x}$$

$$y(x) = (A + B \cdot \ln x) \cdot x^2 //$$

Ⓑ $y(x) = (A + B \ln x) x^2 \Rightarrow y(1) = A = 2$

$$y'(x) = \frac{B}{x} \cdot x^2 + (A + B \ln x) \cdot 2x \Rightarrow y'(1) = B + 2A = 0$$

Therefore; $A = 2, B = -4$

So; $y(x) = (2 - 4 \ln x) x^2 //$

④ $y'' - 2y' + y = e^x + x$

④ For the homogenous solution;

$$y'' - 2y' + y = 0$$

$$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow (\lambda - 1)^2 = 0$$

$\lambda = 1$ with multiplicity 2.

$$\text{So; } y_h(x) = (A + Bx) e^x$$

For $f_1(x) = e^x$;

Derivatives of f_1 have a finite basis: $\{e^x\}$

Since e^x and $x \cdot e^x$ already exist on y_h :

$$y_{p1}(x) = Cx^2 \cdot e^x \Rightarrow y_{p1}'(x) = C \cdot 2x \cdot e^x + Cx^2 \cdot e^x$$

$$y_{p1}''(x) = 2C \cdot e^x + C \cdot 2x \cdot e^x + C \cdot 2x \cdot e^x + Cx^2 \cdot e^x$$

$$2C \cdot e^x + 4C \cdot 2x \cdot e^x + Cx^2 \cdot e^x - 2C \cdot 2x \cdot e^x + 2Cx^2 \cdot e^x + Cx^2 \cdot e^x = e^x$$

$$2C \cdot e^x = e^x \Rightarrow C = \frac{1}{2}$$

$$y_{p1}(x) = \frac{1}{2} x^2 \cdot e^x$$

For $f_2(x) = x$;

Derivatives of f_2 have a finite basis: $\{1, x\}$

$$y_{p2}(x) = D + Ex \Rightarrow y_{p2}'(x) = E, \quad y_{p2}''(x) = 0$$

$$0 - 2E + D + Ex = x \Rightarrow E = 1, \quad D = 2$$

$$y_{p2}(x) = x + 2$$

$$y(x) = y_h(x) + y_{p1}(x) + y_{p2}(x)$$

$$y(x) = A \cdot e^x + Bx \cdot e^x + \frac{1}{2} x^2 \cdot e^x + x + 2 //$$

⑤ $y(x) = A \cdot e^x + Bx \cdot e^x + \frac{1}{2} x^2 \cdot e^x + x + 2 \Rightarrow y(0) = A + 2 = 0$

$$y'(x) = A \cdot e^x + B \cdot e^x + Bx \cdot e^x + x \cdot e^x + \frac{1}{2} x^2 \cdot e^x + 1 \Rightarrow y'(0) = A + B + 1 = 1$$

Therefore; $A = -2, \quad B = 2$

$$\text{So; } y(x) = -2e^x + 2e^x + \frac{1}{2} x^2 \cdot e^x + x + 2 //$$

⑦ $x^2 y'' + x y' - y = \frac{1}{x}$

⑧ For the homogenous solution;

$$x^2 y'' + x y' - y = 0$$

For $y = x^\lambda$; $y' = \lambda x^{\lambda-1}$, $y'' = \lambda(\lambda-1)x^{\lambda-2}$

$$x^2 \cdot \lambda(\lambda-1)x^{\lambda-2} + x \cdot \lambda x^{\lambda-1} - x^\lambda = 0$$

$$x^\lambda (\lambda(\lambda-1) + \lambda - 1) = 0 \Rightarrow \lambda^2 - 1 = 0$$

$$\lambda_1 = 1, \lambda_2 = -1$$

$$y_h(x) = A x^{-1} + B x^1$$

For $y_1 = \frac{1}{x}$, $y_2 = x$ and $f(x) = \frac{1}{x^3}$

$$W = \begin{vmatrix} \frac{1}{x} & x \\ -\frac{1}{x^2} & 1 \end{vmatrix} = \frac{1}{x} + \frac{1}{x} = 2x^{-1}$$

$$W_1 = \begin{vmatrix} 0 & x \\ \frac{1}{x^3} & 1 \end{vmatrix} = 0 - \frac{1}{x^2} = -x^{-2} \Rightarrow C_1' = \frac{W_1}{W} = -\frac{1}{2x} \Rightarrow C_1 = \int -\frac{1}{2x} dx = -\frac{1}{2} \ln|x|$$

$$W_2 = \begin{vmatrix} \frac{1}{x} & 0 \\ -\frac{1}{x^2} & \frac{1}{x^3} \end{vmatrix} = \frac{1}{x^4} - 0 = x^{-4} \Rightarrow C_2' = \frac{W_2}{W} = \frac{x^{-5}}{2} \Rightarrow C_2 = \int \frac{x^{-5}}{2} dx = -\frac{1}{4} x^{-2}$$

$$y_p(x) = -\frac{1}{2} \ln x \cdot \frac{1}{x} - \frac{1}{4} x^{-2} \cdot x = -\frac{\ln x}{2x} - \frac{1}{4x}$$

$$y(x) = y_h(x) + y_p(x) = \frac{A}{x} + Bx - \frac{\ln x}{2x} - \frac{1}{4x}$$

$$y(x) = \frac{A}{x} + Bx - \frac{\ln x}{2x} \Rightarrow y(1) = A + B = 0$$

$$y'(x) = -\frac{A}{x^2} + Bx - \frac{2 - \ln x}{4x^2} \Rightarrow y'(1) = -A + B - \frac{1}{2} = 1$$

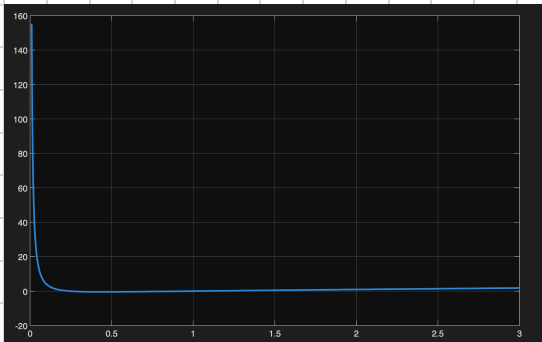
Therefore; $A = -\frac{3}{4}$, $B = \frac{3}{4}$

So; $y(x) = -\frac{3}{4x} + \frac{3x}{4} - \frac{\ln x}{2x}$ //

As $x \rightarrow 0^+$: $-\frac{3}{4x} \rightarrow \infty$ and $-\ln x \rightarrow \infty$. So $y(x) \rightarrow \infty$.

As $x \rightarrow \infty$: $\frac{3x}{4} \rightarrow \infty$ and $\frac{\ln x}{2x} \rightarrow 0$ but $\frac{3x}{4} \rightarrow \infty$. So $y(x) \rightarrow \infty$.

⑥



```
clc;
clear;
close all;

x = linspace(0.01, 3, 1000);
f = -3./(4*x) + (3*x)/4 - log(x)./(2*x);

figure;
plot(x, f, 'LineWidth', 2);
grid on;
```